Open Macroeconomics — Final Exam

A Replication of Uribe and Yue (2006), with an Extension to Brazil, 1996–2026

Celso Nozema   Esthevão Marttioly   Mario Filho   Victor Lucas

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Escola de Economia de São Paulo (EESP)
Open Macroeconomics
Prof. Carlos Eduardo

Agenda

Motivation

- Business cycles in emerging market economies **move closely with the interest rates these countries face in international financial markets**.
  - Low country interest rates $\leftrightarrow$ **economic expansions**.
  - High country interest rates $\leftrightarrow$ **depressed aggregate activity**.
- **What fraction of business-cycle fluctuations in emerging markets is due to movements in the country interest rate?**
- *This question is non-trivial: the country interest rate is not exogenous to the country's own domestic conditions. Spreads respond systematically to output, investment, and the trade balance — **reverse causality is a first-order concern**.*

The Identification Strategy

- To quantify the macroeconomic effects of interest-rate shocks, the first step is to **isolate the exogenous component of country spreads**.
- Following Uribe & Yue (2006), identification combines two ingredients:
  - **Statistical methods** — to identify shocks from the data.
  - **Economic theory** — to assess whether the effects of those identified shocks behave the way a genuine structural shock's effects should.
- *In practice: do the estimated responses to the identified shocks* ***resemble what the theoretical model predicts****?*

Methodology

- **Identify country-spread and US-interest-rate shocks** using a VAR with a recursive ordering. These restrictions cannot be tested using the VAR data alone.
- Hypothesize a theory of the business cycle: **an external benchmark is needed to judge whether those restrictions — and hence the identified shocks — are credible**.
- Impose the shocks found empirically (VAR system) on a theoretical (DSGE) model, then **compare the resulting business-cycle dynamics** (impulse response functions).
- **Verify model validity:** if the two systems generate similar fluctuations in output, investment, and the trade balance under a common shock, this supports treating the empirically identified shocks as genuine structural disturbances.

Methodology: The Algorithm

1. **Estimate the VAR:** link the world rate, the country rate, and domestic variables; identify shocks via a recursive restriction.
2. **Feed the shocks into the DSGE:** impose the VAR-estimated country and world interest rate processes (R and R^{us}) as its exogenous driving forces.
3. **Match the IRFs:** estimate the DSGE's structural frictions (in the style of Christiano, Eichenbaum & Evans, 2001) so the model's theoretical impulse responses best match the VAR's estimated ones.
4. **Compare the dynamics:** impose a common shock in both systems and compare the resulting business-cycle dynamics — VAR-implied vs. DSGE-implied.

Conclusion logic

similar business-cycle dynamics under a common shock $\Rightarrow$ ***the identified shocks are plausible***. *Large, systematic divergence* $\Rightarrow$ ***the identification, the model, or the estimation sample warrants scrutiny***.

An Empirical Model

Variable	Definition
y_t	Real output (log-deviation from a log-linear trend)
i_t	Real investment (log-deviation from a log-linear trend)
tby_t	Trade balance-to-output ratio
R_t^{us}	Gross world (US) real interest rate (log)
R_t	Gross country real interest rate (log)

Full variable construction and data sources: Uribe & Yue (2006), Section 2.1. Replication code reproduces Table 6.1 and the book's own impulse response figures.

Files: statadata.xlsx (baseline 7-country panel) · irs_data_update.mat (16-country, 1994–2012 robustness panel) · uribe_yue_VAR_baseline_v2.mlx

Recursive (Cholesky) Identification

- The contemporaneous matrix A is **lower-triangular with unit diagonal**.
  - Real variables (y, i, tby) react to spread / world-rate innovations **only with a one-quarter lag**.
  - Financial variables (R^{us}, R) react to real shocks **contemporaneously**.
- **R_t^{us} is restricted to a univariate AR(1) process — a small emerging economy cannot move the US interest rate.**
- *Our replication code reproduces Table 6.1 (parameter estimates) and the book's own impulse response figures.*

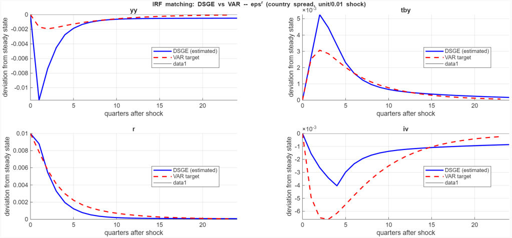
An Open Economy Subject to Interest-Rate Shocks

1. **Time-to-build capital:** Investment projects take 4 quarters to complete, plus convex capital-adjustment costs $\Rightarrow$ smooth, hump-shaped investment dynamics.
2. **External habit formation** ("catching up with the Joneses"): prevents excessive consumption volatility.
3. **Working-capital-in-advance constraint:** firms must finance a fraction η of the wage bill in advance $\Rightarrow$ a rate hike raises firms' effective labor cost directly, producing a genuine supply-side contraction.
4. **Information timing:** production and absorption decisions are made one period before that period's interest-rate shock is realized — mirrors the VAR's own identifying assumption exactly.

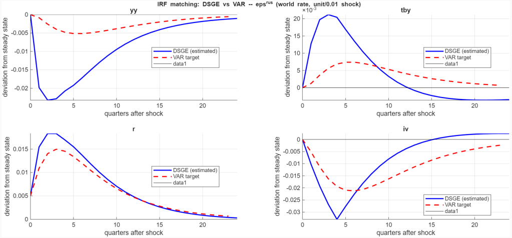
Replication of the Baseline Model

- Full equations and derivations can be found in Uribe & Yue (2006), Schmitt-Grohé & Uribe (2017, Ch. 6), and our own derivation document (`uribe_yue_derivation`) — which **translates every resulting equation into Dynare code** (`DSGE_uribe_yue_IRFmatch.mod`).
- Results: the two sets of business-cycle dynamics — VAR-implied and DSGE-implied, both under the same shock — **are close in this baseline case, confirming the findings of the original study**.
- **Conclusion:** this supports treating the identified country-spread and US-rate shocks as economically plausible — the theory, taken seriously, reproduces the dynamic effects the data-driven VAR attributes to these shocks.

Baseline IRF Comparison (7-Country Panel)



Country-spread shock: VAR-estimated vs. DSGE-implied.



World (US) rate shock: VAR-estimated vs. DSGE-implied.

VAR-implied and DSGE-implied dynamics are close, confirming Uribe & Yue (2006).

Extending the Model to Brazil Only, 1996–2026

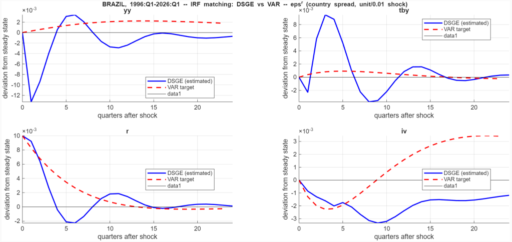
- While panel data provides a larger effective sample and more efficient estimates, it also **imposes a homogeneity assumption on the dynamic (slope) coefficients**: Argentina, Brazil, Ecuador, etc. are assumed to share the same estimated responses to shocks — even though country-specific intercepts are accounted for via fixed effects.
- A single-country sample over a much longer horizon relaxes cross-country homogeneity entirely, but now **requires coefficient stability over a much longer, more heterogeneous period** — difficult to defend across the GFC, COVID, and the 2021–23 hiking cycle.
- Idiosyncratic events affecting Brazil during this period (e.g., political turmoil) are also **a likely source of regime changes**.

Files: uribe_yue_VAR_Brazil.mlx · DSGE_uribe_yue_IRFmatch_Brazil.mod · replicate_IRF_uribe_yue_matching_Brazil.m

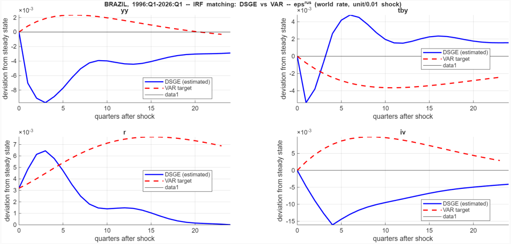
Analysing Results for Brazil

- *Many caveats apply before treating these results as conclusive.*
- The model — calibrated as in Uribe & Yue (2006) — **does not validate well when looking at Brazil-only data**, a signal that is masked when pooled into the 7-country panel.
- **Higher persistence of R^{us}:** in the original paper, $\rho \approx 0.83$. Once extended to 2026 — which now spans the post-GFC near-zero-rate period — persistence is $\rho \approx 0.96$, which dramatically slows the estimated IRFs' convergence back to steady state.
- This pattern confirms one of the robustness checks in Schmitt-Grohé & Uribe (2017): Section 6.2.1 extended the sample through 2012, and likewise **found significantly more persistent estimated responses** than the original 1994–2001 baseline.

Brazil IRF Comparison



Country-spread shock: VAR-estimated vs. DSGE-implied.



World (US) rate shock: VAR-estimated vs. DSGE-implied.

VAR and DSGE dynamics diverge sharply — especially the response of output.

Interpreting the Divergence

- **VAR and DSGE business-cycle dynamics diverge sharply** — especially the response of output, which stays economically implausible even after re-estimating the model's frictions.
- We also tested relaxing the VAR's "same-quarter exogeneity" identifying assumption; **the divergence got worse, not better** — by the paper's own logic, this argues against identification being the culprit.
- More likely explanation: **omitted common drivers** (e.g., commodity prices / terms of trade) may move spreads and output together, contaminating the identified "shocks."
- **Takeaway:** a genuine, informative external-validity failure — this model, calibrated as in Uribe & Yue (2006), does not straightforwardly extend to Brazil over this long, heterogeneous sample.

Conclusion

- The baseline replication of Uribe & Yue (2006) succeeds: **VAR-implied and DSGE-implied business-cycle dynamics closely match** across the 7-country panel, under both country-spread and world-rate shocks.
- Extending the same, identically calibrated model to Brazil alone (1996–2026) **exposes a genuine external-validity failure**: output dynamics diverge sharply between the two systems.
- The divergence does not appear to be an identification problem — **relaxing the VAR's identifying assumption makes it worse, not better**.
- The most likely explanation is **omitted common drivers**, such as commodity prices or terms of trade, which may move spreads and output together and contaminate the identified shocks.

References

Christiano, L. J., Eichenbaum, M., & Evans, C. L. (2001). *Nominal Rigidities and the Dynamic Effects of a Shock to Monetary Policy.* NBER Working Paper No. 8403.

Mendoza, E. G. (1991). *Real Business Cycles in a Small Open Economy.* American Economic Review, 81(4), 797–818.

Schmitt-Grohé, S., & Uribe, M. (2017). *Open Economy Macroeconomics.* Princeton University Press.

Uribe, M., & Yue, V. Z. (2006). *Country Spreads and Emerging Countries: Who Drives Whom?* Journal of International Economics, 69(1), 6–36.